Project Documentation

Student Performance Analytics System

Course: Python with AI
Organization: EWB Courses
Submitted by: Sachin Kumar
Technology: Python, NumPy, Pandas
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1. Project Title

Student Performance Analytics System

The Student Performance Analytics System is a Python-based mini-project designed to process and analyze student academic performance using marks and attendance data. It calculates total marks, average marks, grades, pass/fail results, subject-wise statistics, and identifies the top-performing students.

2. Objective

The main objective of this project is to develop a simple data analysis system that helps evaluate student performance efficiently.

The project aims to:

- Read and organize student records from a CSV file.
- Calculate total marks and average marks for each student.
- Assign grades based on average marks.
- Determine pass/fail status using subject marks and attendance.
- Analyze the highest, lowest, and average marks for each subject.
- Calculate overall class statistics and identify the top three academic performers.

This project demonstrates how Python and its data analysis libraries can be used to convert raw student records into meaningful information.

3. Technologies Used

Python: Used to implement the program logic, functions, loops, conditional statements, and data processing.

Pandas: Used to read the CSV dataset, organize records in a DataFrame, filter student records, and display analysis results.

NumPy: Used for numerical calculations, including total marks, averages, and subject-wise statistical analysis.

CSV (Comma-Separated Values): Used as the data storage format for student information, subject marks, and attendance.

Visual Studio Code: Used as the development environment to write, edit, and execute the Python program.

4. Dataset Description

The project uses a CSV file named students.csv, which contains records of 20 students. The dataset stores student details, marks obtained in five subjects, and attendance percentage. It is loaded into a Pandas DataFrame for processing and analysis.

The dataset contains the following columns:

1. Student_ID: A unique identification number assigned to each student.
2. Name: The name of the student.
3. Python: Marks obtained by the student in Python.

The marks for each subject and the attendance percentage are recorded on a scale of 0 to 100. The dataset is used to calculate total marks, average marks, grades, pass/fail results, subject-wise statistics, and the top three academic performers.

5. Implementation

The project is organized into three main files: main.py, functions.py, and students.csv.

Step 1: Loading the Dataset

The Pandas library reads the student records from students.csv using the read_csv() function. The records are stored in a DataFrame and displayed before analysis.

Step 2: Calculating Total and Average Marks

The calculate_totals_and_averages() function extracts subject marks using Pandas and NumPy.

- Total marks are calculated by adding marks across all five subjects.
- Average marks are calculated by dividing the total marks by the number of subjects.
- The calculated values are stored in the DataFrame.

Step 3: Assigning Grades

The assign_grades() function uses conditional statements to assign grades based on each student's average marks.

Average Marks	Grade
90 and above	O
80–89.99	E
70–79.99	A
60–69.99	B
50–59.99	C
40-49.99	D
Below 40	F

Step 4: Evaluating Pass/Fail Results

The evaluate_results() function uses loops and conditional statements to evaluate student results.

A student passes only when all subject marks are at least 40 and attendance is at least 75%. Otherwise, the student is marked as failed.

Therefore, a student can achieve a high grade but still fail because of insufficient attendance or a subject mark below the passing requirement.

Step 5: Performing Subject-Wise Analysis

The get_subject_analysis() function calculates the average, highest, and lowest marks for each subject using NumPy operations.

Step 6: Generating Class Statistics

The main program calculates the total number of students processed, overall class average, number of students who passed, and number of students who failed.

Step 7: Identifying Top Performers

The program sorts students by total marks in descending order and displays the top three academic performers.

The leaderboard is based on total marks, independently of pass/fail status. Therefore, a student who fails due to attendance can still appear among the top three performers.

6. Key Features

The major features of the Student Performance Analytics System are:

1. **CSV Data Loading:** Reads student records from an external CSV file.
2. **Automated Calculations:** Calculates total and average marks for each student.
3. **Grade Assignment:** Assigns grades using predefined average-mark ranges.
4. **Pass/Fail Evaluation:** Checks subject passing marks and minimum attendance.
5. **Class Performance Summary:** Displays the number of students, overall class average, and pass/fail counts.
6. **Subject-Wise Analysis:** Reports average, highest, and lowest marks for each subject.

7. **Top Three Performers:** Ranks students according to their total marks.
8. **Modular Code Structure:** Separates the main program from reusable functions.

7. Output / Screenshots

The program produces a report containing student records, individual performance, class statistics, subject-wise analysis, and the top three academic performers.

Screenshot 1: Student Dataset

```
PS D:\sachin project\students_performance_analytics> python -u "d:\sachin project\students_performance_analytics\main.py"
STUDENT PERFORMANCE ANALYTICS SYSTEM
**************************************
==========STUDENT DATA==========
 Student_ID      Name  Python  SQL  HTML  CSS  JavaScript  Attendance
        101    Anjali      85   78    85   95          92          92
        102     Sanvi      72   80    78   82          88          85
        103    Saniya      95   92    50   40          70          75
        104     Sneha      60   65    91   94          96          94
        105     Arjun      88   85    65   70          81          68
        106     Rohan      35   42    60   55          76          90
        107     Sohan      78   82    80   88          89          95
        108   Shubham      90   88    92   91          94          80
        109     Sayim      55   60    58   62          82          65
        110     Mikki      30   38    35   45          65          89
        111     Omair      82   85    87   90          91          84
        112   Raushan      70   74    70   78          85          91
        113    Ritwik      48   70    74   71          80          76
        114    Sachin      91   50    50   56          78          96
        115   Ruchika      77   94    89   96          95          69
        116 Dipanwita      79   52    85   84          87          55
        117  Satarupa      94   42    45   40          72          47
        118   Vaibhav      58   83    81   86          90          93
        119     Anmol      60   62    93   92          97          39
        120      Bitu      50   66    60   65          83          57
```

Screenshot 2: Student Performance Report

```
**********STUDENT PERFORMANCE**********

    Student_ID      Name  Total  Average Grade Result
0          101     Anjali    435     87.0     E   Pass
1          102      Sanvi    400     80.0     E   Pass
2          103     Saniya    347     69.4     B   Pass
3          104      Sneha    406     81.2     E   Pass
4          105      Arjun    389     77.8     A   Fail
5          106      Rohan    268     53.6     C   Fail
6          107      Sohan    417     83.4     E   Pass
7          108    Shubham    455     91.0     O   Pass
8          109      Sayim    317     63.4     B   Fail
9          110      Mikki    213     42.6     D   Fail
10         111      Omair    435     87.0     E   Pass
11         112    Raushan    377     75.4     A   Pass
12         113     Ritwik    343     68.6     B   Pass
13         114     Sachin    325     65.0     B   Pass
14         115    Ruchika    451     90.2     O   Fail
15         116  Dipanwita    387     77.4     A   Fail
16         117   Satarupa    293     58.6     C   Fail
17         118    Vaibhav    398     79.6     A   Pass
18         119      Anmol    404     80.8     E   Fail
19         120       Bitu    324     64.8     B   Fail
```

Screenshot 3: Class and Subject-Wise Analysis

```
**********SUBJECT ANALYSIS**********

Total Students Processed: 20
Overall Class Average: 73.84
Number Of Students Passed: 11
Number of Students Failed: 9

---------Subject-Wise Analysis-------
            Average  Highest  Lowest
Python        69.85     95.0    30.0
SQL           69.40     94.0    38.0
HTML          71.40     93.0    35.0
CSS           74.00     96.0    40.0
JavaScript    84.55     97.0    65.0
```

Screenshot 4: Top Three Academic Performers

```
--------Top 3 Performers--------
  Student_ID    Name  Total  Average Grade
         108 Shubham    455     91.0     O
         115 Ruchika    451     90.2     O
         101  Anjali    435     87.0     E
```

8. Final Outcome

The Student Performance Analytics System successfully demonstrates how Python, NumPy, and Pandas can be used to analyze student academic records.

The system processes the student dataset, calculates total and average marks, assigns grades, evaluates results based on marks and attendance, and generates class-level and subject-level statistics.

It also identifies the top three students based on total marks. The project provides a simple and organized way to understand student performance while demonstrating fundamental programming and data analysis concepts.

9. Challenges & Learning

Challenges

During the development and testing process, attention was required to ensure that the CSV file was loaded correctly, the Python files were saved properly, and the output was displayed in the expected format.

Another important aspect was implementing the pass/fail condition correctly so that both subject marks and attendance were considered.

Learning Outcomes

Through this project, I learned:

- How to read and process CSV files using Pandas.
- How to perform numerical calculations using NumPy.
- How to create reusable functions in Python.

- How to use loops and conditional statements to implement decision-making logic.
- How to calculate descriptive statistics and analyze data.
- How to organize a Python project into multiple files.
- How to test a program and interpret its output.

This project strengthened my understanding of Python fundamentals and introduced me to practical data analysis.

10. GitHub Repository

Repository Name: student-performance-analytics

Repository URL: https://github.com/SachinKumar-IT/student-performance-analytics